NISARGA MURTHY
CYBERSECURITY ENGINEER

nisargamurthybp@gmail.com | +91-6360020411 | Bengaluru, Karnataka, India
GitHub: github.com/nxshh | LinkedIn: linkedin.com/in/nisargamurthy

PROFESSIONAL SUMMARY

Cybersecurity Engineer with hands-on experience in penetration testing, vulnerability assessment, threat modeling, and security research. Proficient in offensive security tools including Kali Linux, Metasploit, Burp Suite, Nmap, and Wireshark, with all findings mapped to OWASP Top 10 and CVE frameworks. Published researcher in ML-based threat detection, combining automated attack simulation with machine learning to identify honeyfile evasion techniques. Additional background in secure software development — building and hardening REST APIs, implementing RBAC, JWT authentication, and cloud security controls on AWS. Holds industry certifications from Cisco, EC-Council, and Pearson. Actively pursuing advanced security certifications.

TECHNICAL SKILLS

Offensive Security: Penetration Testing, Vulnerability Assessment, Exploitation, Red Team Operations, OWASP Top 10
Security Tools: Kali Linux, Metasploit Framework, Burp Suite, Nmap, Wireshark, Nikto, Netcat
Attack Simulation: Metasploitable 2, CVE exploitation, network scanning, enumeration, privilege escalation
Threat Detection & Analysis: ML-based anomaly detection, threat modeling, risk analysis, IOC identification, log analysis
Digital Forensics: Evidence analysis, forensic investigation fundamentals, chain of custody, artifact extraction
Network Security: TCP/IP, network traffic analysis, firewall rules, IDS/IPS concepts, VPN, network hardening
Secure Development (SSDLC): Secure REST API design, JWT, RBAC, input validation, OWASP secure coding, AWS IAM hardening
Cloud Security: AWS (EC2, IAM, S3) — least-privilege access, secure environment configuration, access controls
Programming & Scripting: Python, Shell Scripting, JavaScript (Node.js) — security automation and tooling
Machine Learning (Security): Random Forest, SVM, Autoencoders — anomaly detection, feature engineering, model evaluation
Platforms & Environments: Linux (Kali, Ubuntu), VMware, VirtualBox, Docker
Frameworks & Standards: OWASP Top 10, MITRE ATT&CK (familiar), NIST Cybersecurity Framework (familiar), CVE

PROFESSIONAL EXPERIENCE

Junior Cyber Security Analyst *Jul 2023 – Aug 2023*
Verzeo | Bengaluru, India

- Performed structured penetration tests across 4 lab environments using Kali Linux, Metasploit, Nmap, and Burp Suite; all findings documented and mapped to OWASP Top 10 vulnerability classes.
- Identified 8 critical system misconfigurations including exposed services, weak credentials, and unpatched CVEs across Linux hosts; mitigation recommendations adopted into team security runbooks.
- Conducted network traffic analysis using Wireshark to detect anomalous patterns, unauthorized connections, and potential data exfiltration indicators.
- Assisted in exploitation, risk analysis, and mitigation planning — supporting the full vulnerability management lifecycle from discovery through remediation tracking.
- Performed Linux system hardening tasks including service minimization, permission auditing, and firewall rule review to reduce attack surface.

Backend Security-Focused Developer Intern *Jul 2025 – Dec 2025*
Basshub Locateus Pvt Ltd | Bengaluru, India

- Designed and implemented secure REST APIs applying OWASP secure coding practices end-to-end — JWT authentication, role-based access control (RBAC), and strict input validation closing 5 OWASP Top-10 class vulnerabilities.
- Configured AWS infrastructure (EC2, IAM, S3) with least-privilege IAM policies, enforcing the principle of least privilege and eliminating overprivileged service accounts across staging and production.
- Applied secure SDLC practices throughout development: threat modeling at design, security-focused code review, and integration tests covering authentication bypass and injection attack scenarios.
- Implemented Docker container security best practices — non-root container users, environment secret management, and secure CI/CD pipeline configurations in GitHub Actions.
- Reduced API response latency by 30% through query optimization while maintaining all security controls — demonstrating ability to balance performance and security requirements.

PROJECTS

Breaking the Bait: ML-Based Honeyfile Detection System (Published) *2025*

- Designed and built a fully automated attack simulation environment on Kali Linux and Metasploitable 2, generating a labeled dataset of 1,000+ real and decoy file system entries with realistic metadata to simulate adversarial conditions.
- Applied feature engineering and trained three ML models — Random Forest, SVM, and Autoencoder — to detect honeyfile evasion techniques; Random Forest achieved 94% F1-score under adversarial test conditions.
- Implemented rigorous evaluation methodology: stratified k-fold cross-validation, feature importance analysis, confusion-matrix breakdown — simulating detection under real red team engagement conditions.
- Published research and open-sourced complete implementation on GitHub; system designed to integrate as a post-breach detection layer in SOC workflows with no prior labeling of the target environment required.
- Outcome: demonstrates applied capability in threat detection engineering, attack simulation, machine learning for security, and red team tool awareness.

TrippCard OTA — Secure Cloud-Native API Platform *2025*

- Built and hardened a production REST API backend (Node.js + PostgreSQL) applying OWASP Top 10 secure coding guidelines across all 12+ endpoints — authentication, authorization, injection prevention, and error handling.
- Implemented JWT-based authentication, RBAC, and input sanitization, preventing unauthorized access and injection attacks; passed internal security review with zero critical findings post-hardening.
- Configured AWS (EC2, IAM, S3) with least-privilege access controls and secured Docker containerization with environment secret management — demonstrating end-to-end secure deployment practices.

EDUCATION

Master of Computer Applications (MCA) 2025
Reva University, Bengaluru, India

Bachelor of Computer Applications (BCA) 2023
Amrita Vishwa Vidyapeetham, Mysore, India

CERTIFICATIONS

- Cisco Certified Support Technician – Cybersecurity (CCST) | Cisco
- IT Specialist – Network Security | Pearson
- Digital Forensics Essentials | EC-Council
- DevOps Advanced Projects with AI | Udemy
- AWS Certified DevOps Engineer – Professional (In Progress) | Amazon Web Services